SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.

Smart Insurance Policy Recommendation and Claim Management System

Course: CSA0905- Java Programming

Guide:
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Professor

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Problem Background

- The project belongs to the **Insurance Technology (InsurTech)** domain and uses AI to provide smart insurance policy recommendations and efficient claim management.
- Customers often find it difficult to compare multiple insurance policies, while traditional claim processing is slow and paper-based.
- Existing systems lack personalized recommendations, resulting in poor customer experience and delayed claim settlements.
- The proposed system automates policy recommendations and claim processing, reducing manual effort, errors, and processing time.
- The system is designed for **customers, insurance companies, agents, claim officers, and policy administrators**, ensuring faster, secure, and transparent insurance services.

Problem Statement

- Customers face difficulties in selecting suitable insurance policies, while traditional claim processing is slow, complex, and prone to manual errors.
- According to industry reports, **insurance claim processing can take several days or even weeks**, and many customers struggle to understand and compare available policy options.
- The proposed AI-based system recommends the most suitable insurance policy, automates claim processing, and enables secure real-time claim status tracking.
- The expected impact is **faster claim settlements, reduced operational costs, improved customer satisfaction, increased transparency, and better decision-making** for both customers and insurance providers.

Objectives

Primary Objective: To develop a **Smart Insurance Policy Recommendation and Claim Management System** that provides personalized policy suggestions and automates claim processing for faster, secure, and transparent insurance services.

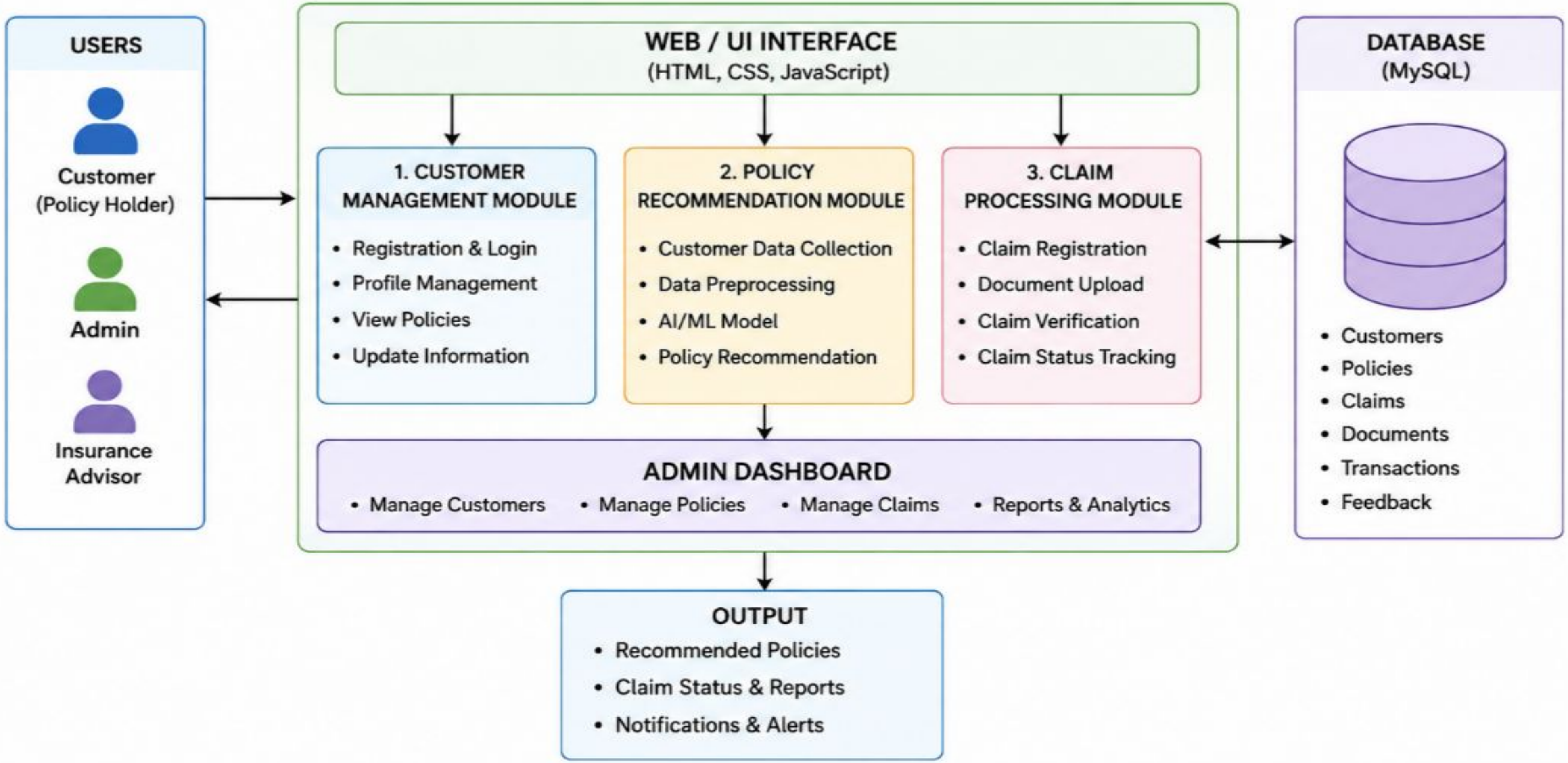
Measurable Objectives

- Recommend the most suitable insurance policy based on customer profile and preferences.
- Reduce claim processing time through automated verification and online claim submission.
- Improve recommendation accuracy using AI-based analysis.
- Enable real-time claim status tracking and customer notifications.
- Ensure secure storage and management of customer and policy data

Literature Survey

Title of the Paper	Authors and Years	Contribution	Limitations
Learning Recommendations from User Actions in the Item-poor Insurance Domain	Simone Borg Bruun, Maria Maistro, Christina Lioma (2022)	Improved insurance policy recommendation using user behavior.	Requires sufficient user interaction data.
Recommending Insurance Products by Using Users' Sentiments	Rohan Parasrampuria, Ayan Ghosh, Suchandra Dutta, Dhrubasish Sarkar (2021)	Used sentiment analysis for better policy recommendations.	Depends on quality of customer reviews.
DCDIR: A Deep Cross-Domain Recommendation System for Cold Start Users in Insurance Domain	Ye Bi, Liqiang Song, Mengqiu Yao, Zhenyu Wu, Jianming Wang, Jing Xiao (2020)	Recommended policies for new customers using deep learning.	Needs large training datasets.
Deep Claim: Payer Response Prediction from Claims Data with Deep Learning	Byung-Hak Kim, Seshadri Sridharan, Andy Atwal, Varun Ganapathi (2020)	Predicted claim approval using deep learning.	Limited to claim prediction only.
Recommending Target Actions Outside Sessions in the Data-poor Insurance Domain	Simone Borg Bruun, Christina Lioma, Maria Maistro (2024)	Improved recommendations in data-poor insurance systems	Does not support complete claim management.

System


USERS
Customer
(Policy Holder)
Admin
Insurance
Advisor
WEB / UI INTERFACE
(HTML, CSS, JavaScript)
1. CUSTOMER
MANAGEMENT MODULE
• Registration & Login
• Profile Management
• View Policies
• Update Information
2. POLICY
RECOMMENDATION MODULE
• Customer Data Collection
• Data Preprocessing
• AI/ML Model
• Policy Recommendation
3. CLAIM
PROCESSING MODULE
• Claim Registration
• Document Upload
• Claim Verification
• Claim Status Tracking
ADMIN DASHBOARD
• Manage Customers
• Manage Policies
• Manage Claims
• Reports & Analytics
DATABASE
(MySQL)
• Customers
• Policies
• Claims
• Documents
• Transactions
• Feedback
OUTPUT
• Recommended Policies
• Claim Status & Reports
• Notifications & Alerts

Existing vs Proposed Systems

Key Features of the Existing System

- Manual customer registration and record management.
- Generic policy recommendations for all customers.
- Paper-based documentation and record keeping.
- Manual claim submission and verification.
- Separate databases with less data integration.
- Limited customer support and tracking.

Key Features of the Proposed System

- Digital customer registration and profile management.
- Automated claim submission and processing.
- Real-time claim status tracking.
- Centralized and secure customer database.
- Reduced paperwork through digital records.

Requirements

- The system supports customer registration, profile management, AI-based policy recommendation, claim submission, and claim tracking with a secure and user-friendly interface.
- It requires hardware such as an Intel Core i3/i5 processor, 8 GB RAM, 500 GB storage, and a stable internet connection for smooth operation.
- The software environment includes Windows 10/11, Java, MySQL, Eclipse IDE, Apache Tomcat, and Google Chrome for development and deployment.
- The system uses customer, insurance policy, and claim datasets to provide accurate policy recommendations and efficient claim processing.
- Notification APIs such as Email/SMS and Payment Gateway APIs are integrated to improve communication, premium payments, and overall system functionality.

Planning

Phase 1: Requirement Analysis

Problem identification
Literature survey
Requirement gathering

Phase 2: System Design

System architecture
Database design
UI/UX design

Phase 3: Development

Customer Management Module
Policy Recommendation Module
Claim Processing Module

Phase 4: Testing

Unit testing
Integration testing
System testing and bug fixing

Phase 5: Deployment

System deployment
User training
Documentation

Expected Outcomes

Deliverables

- Customer Management Module for managing user details.
- AI-based Policy Recommendation Module.
- Claim Processing Module with status tracking.
- Admin Dashboard for monitoring the system.
- Project report, database, and source code..

Benefits

- Provides personalized insurance policy recommendations.
- Reduces manual work and paperwork.
- Speeds up claim processing and approval.
- Improves customer satisfaction and service quality.
- Ensures secure and efficient data management.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